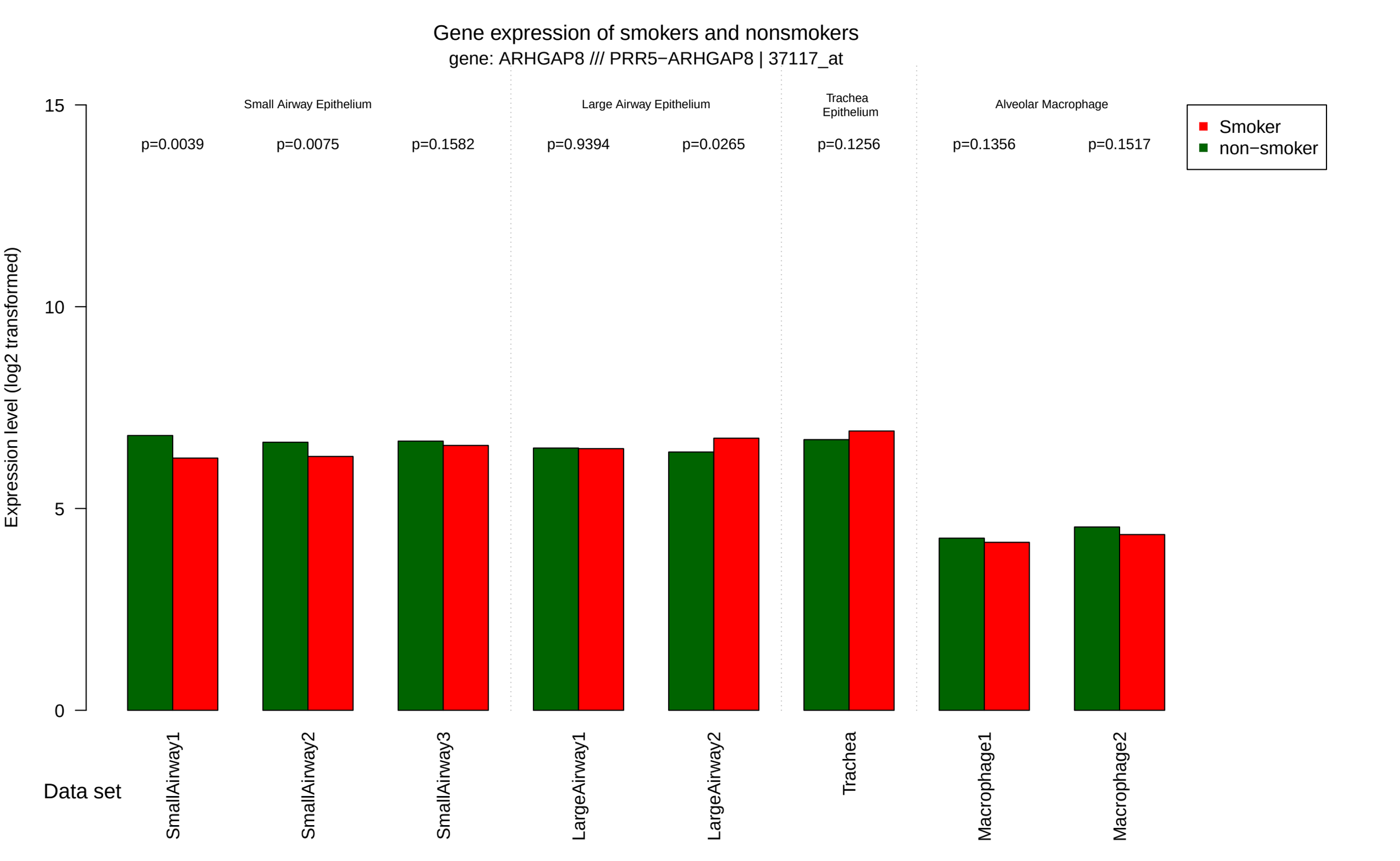
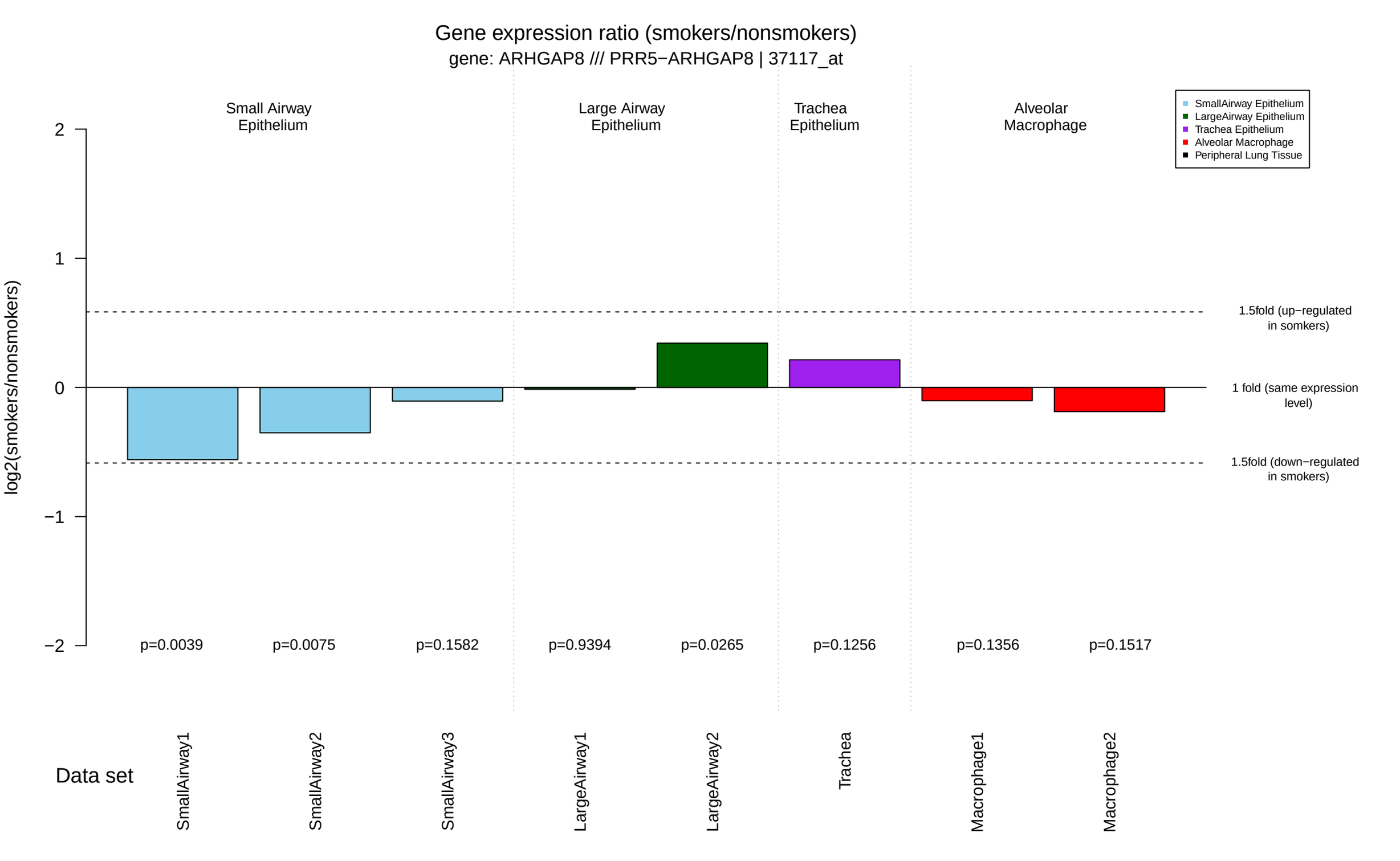
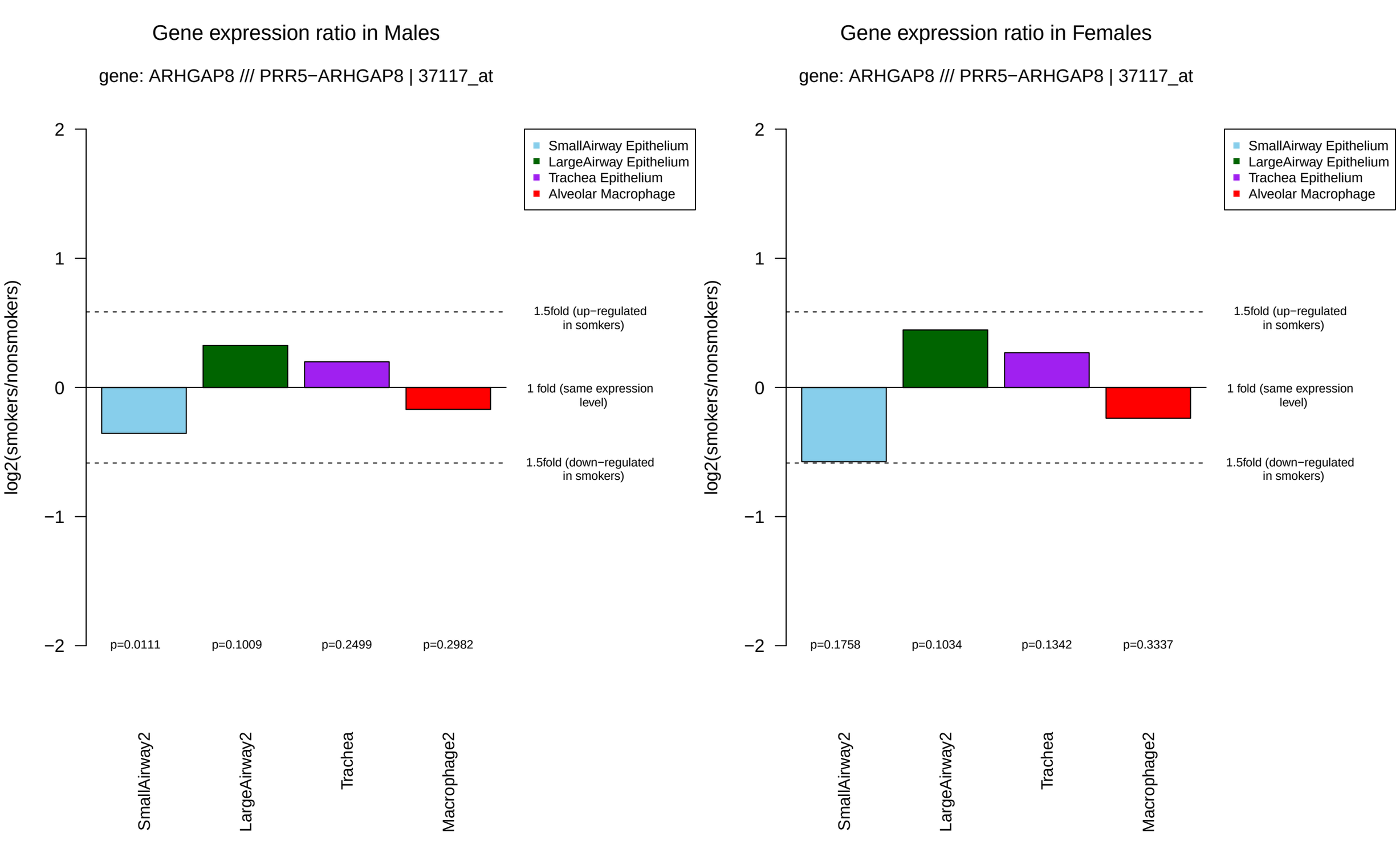
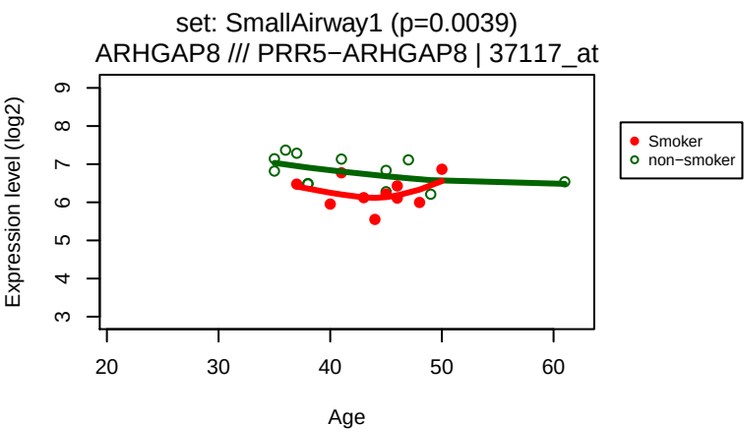
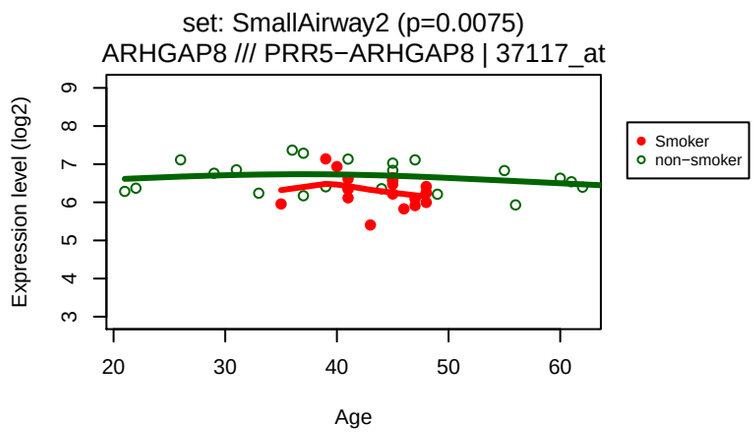
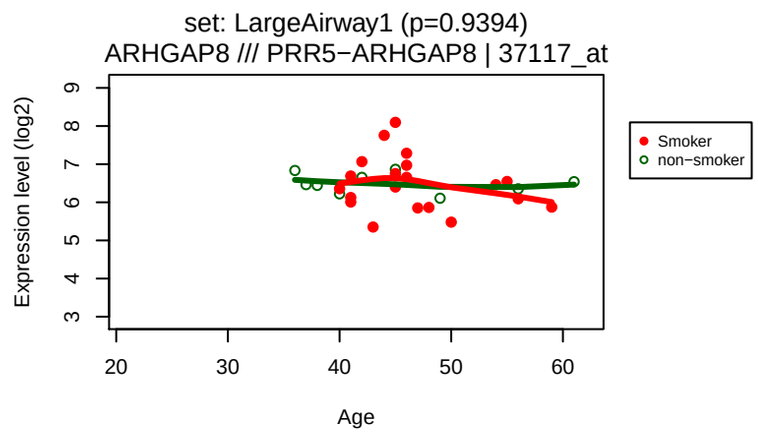

Gene expression of smokers and nonsmokers
gene: ARHGAP8 /// PRR5−ARHGAP8 | 37117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0039
p=0.0075
p=0.1582
p=0.9394
p=0.0265
p=0.1256
p=0.1356
p=0.1517
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

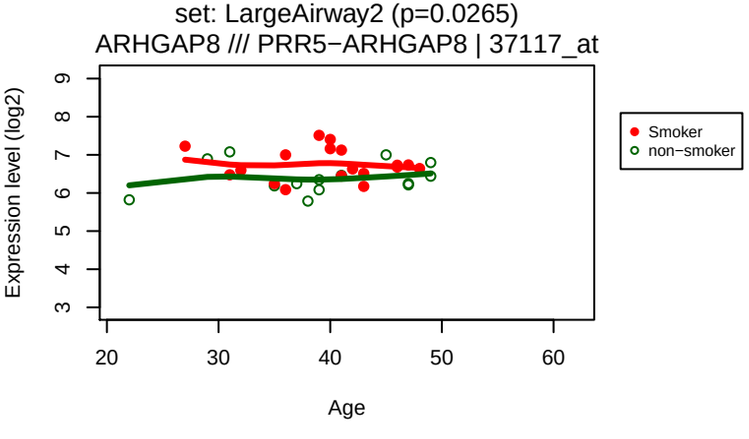

Gene expression ratio (smokers/nonsmokers)
gene: ARHGAP8 /// PRR5−ARHGAP8 | 37117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0039
p=0.0075
p=0.1582
p=0.9394
p=0.0265
p=0.1256
p=0.1356
p=0.1517
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

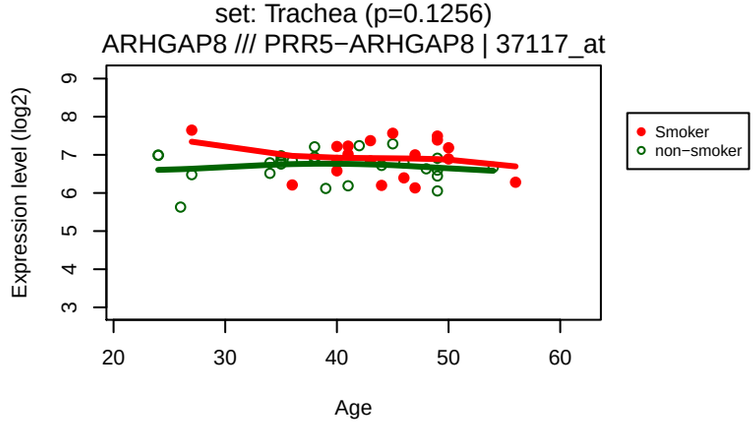

Gene expression ratio in Males
gene: ARHGAP8 /// PRR5−ARHGAP8 | 37117_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0111
p=0.1009
p=0.2499
p=0.2982
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGAP8 /// PRR5−ARHGAP8 | 37117_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1758
p=0.1034
p=0.1342
p=0.3337
SmallAirway2
LargeAirway2
Trachea
Macrophage2

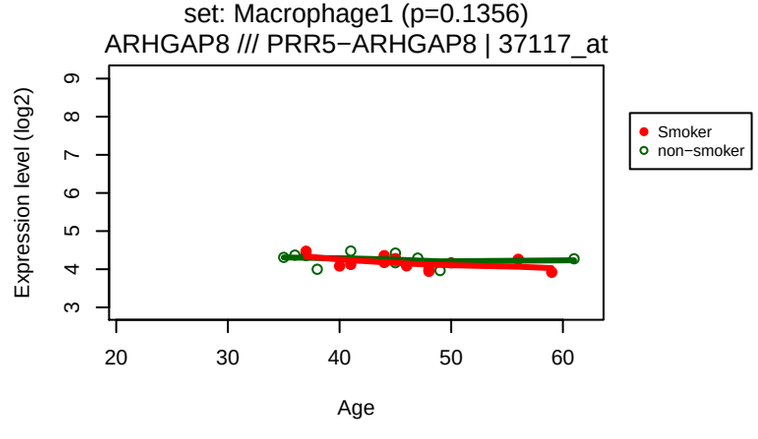

set: SmallAirway1 (p=0.0039)
ARHGAP8 /// PRR5−ARHGAP8 | 37117_at
Expression level (log2)
Age
Smoker
non−smoker

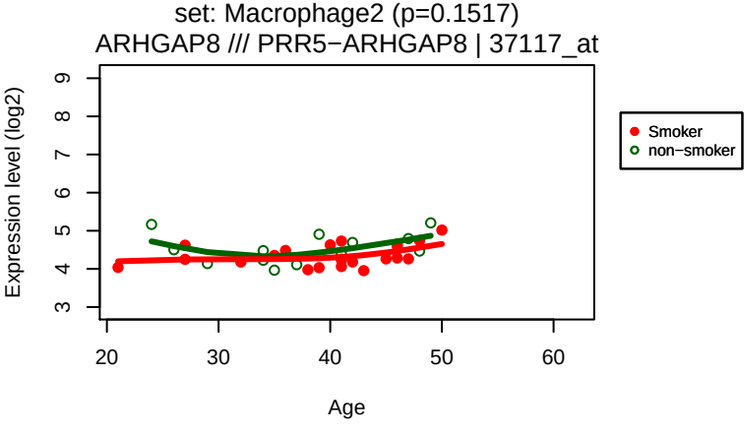

set: SmallAirway2 (p=0.0075)
ARHGAP8 /// PRR5−ARHGAP8 | 37117_at
Expression level (log2)
Age
Smoker
non−smoker

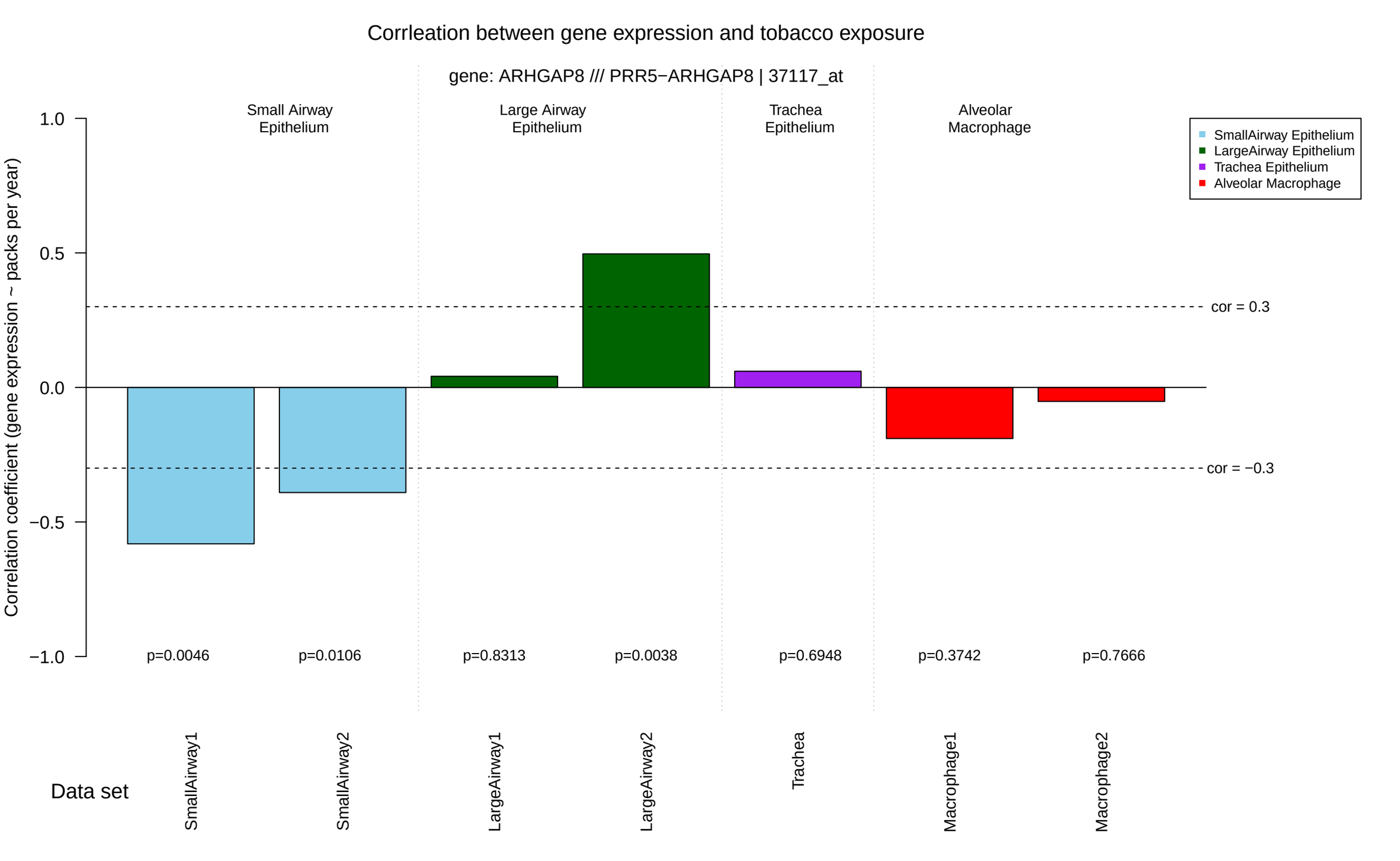

set: LargeAirway1 (p=0.9394)
ARHGAP8 /// PRR5−ARHGAP8 | 37117_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0265)
ARHGAP8 /// PRR5−ARHGAP8 | 37117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1256)
ARHGAP8 /// PRR5−ARHGAP8 | 37117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1356)
ARHGAP8 /// PRR5−ARHGAP8 | 37117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1517)
ARHGAP8 /// PRR5−ARHGAP8 | 37117_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGAP8 /// PRR5−ARHGAP8 | 37117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0046
p=0.0106
p=0.8313
p=0.0038
p=0.6948
p=0.3742
p=0.7666
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2